MIL-C-14460C
14 December 1981
SUPERSEDING
MIL-C-14460B
22 March 1966

MILITARY SPECIFICATION

CORROSION REMOVING COMPOUND, SODIUM HYDROXIDE BASE; FOR ELECTROLYTIC OR IMMERSION APPLICATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers corrosion removing compound, containing sodium hydroxide plus a water-soluble chelate compound or a sequestrant compound, and with or without alkali metal cyanide. Solutions of these compounds are suitable for rust removal, either by simple immersion of the articles or by electrolytic action, without causing material change in the dimensional characteristics of the treated articles (see 6.1).

1.2 Classification. Corrosion removing compounds covered by this specification shall be of the following types, as specified (see 6.2):

Type I - Sodium hydroxide plus chelate compounds or sequestrant compounds (see 6.3).
Type II - Sodium hydroxide-sodium cyanide mixture plus a chelate compound or a sequestrant compound.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

FEDERAL

PPP-D-729 - Drums; Metal, 55-gallon (for Shipment of Noncorrosive Materials)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SSS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

/PSC 6850/

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.

Table I. Composition.

Ingredient	Requirement (percent by weight) Type I	Type II	Test Paragraph
Sodium hydroxide	54.0 max	35.0 min	4.5.4
1/Trisodium salt of N-hydroxyethyl-ethylenediaminetriacetic acid (3Na·EDTAOH)	13.0 min	---	4.5.7
Chelate or sequestrant compound (EDTA tetrasodium salt dihydrate)	---	25 to 35	4.5.9
Sodium gluconate	25.0 min	---	4.5.6
Sodium cyanide	---	25 to 35	4.5.8
Phosphorus and silica, total as PO_4 plus SiO_2	0.25 max	0.25 max	4.5.5
Others, including foamers	7.0 max	4.0 max	---

1/N-hydroxyethylethylenediaminetriacetic acid may be used instead of the trisodium salt if the amount of acid when calculated as the trisodium salt meets the requirements of this specification.

3.3 Solubility (turbidity). The corrosion removing compound shall be completely soluble in water when tested as specified in 4.5.2.

3.4 Foaming characteristics. When tested as in 4.5.3, the corrosion removing compound shall show a foam head of between 1/8 and 1/4 inch.

3.5 Etching characteristics (type II only). When tested as in 4.5.10, the steel plates shall show no evidence of etching.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.

4.5.2 Solubility (turbidity).

4.5.2.1 Type I. Examine the sample solution prepared in 4.5.1.1 for turbidity and for conformance with 3.3.

4.5.2.2 Type II. Heat 500 mL of the sample solution from 4.5.1.2 to 120°F ± 10 (49°C ± 5.6) and examine for turbidity.

4.5.3 Foaming characteristics.

4.5.3.1 Preparation of steel panels. The panels, each measuring 1 X 4 X 1/8 inches (25 X 100 X 3.175 mm), shall be SAE 1020 cold-rolled steel. A 3/16-inch (4.8 mm) hole shall be drilled in one corner of each panel for attaching power leads by means of a steel bolt and nut. The surfaces of the panels shall be abraded to a surface roughness not greater than 32 microinch (0.8 µm).

4.5.3.2 Test procedure. Place 500 mL of the sample solution (see 4.5.1.1 and 4.5.1.2) into a 600 mL standard form beaker. Place the two panels (see 4.5.3.1) on opposite sides in the beaker, immersing them to a depth of 2 inches (50 mm). Attach a suitable direct current source, making one panel anodic (positive) and the other cathodic (negative). (A 6-volt storage battery may be used as the supply source.) Apply a current of 4 amperes and hold steady for 30 minutes at a temperature between 80°F (27°C) and 130°F (54°C) (by means of a water bath, if necessary). After the current has been off for 2 minutes, the height of foam at the center of the beaker shall be measured. Reserve the panels for the test of 4.5.10.

4.5.4 Sodium hydroxide content.

4.5.4.1 Preparation of color blank. Immediately before performing the test for sodium hydroxide content, prepare a color blank by dissolving 1 g of sodium cyanide in 80 mL distilled water in a 125 mL Erlenmeyer flask and then adding 16 drops of LaMotte sulfo-orange indicator.

4.5.4.2 Preparation of 1N hydrochloric acid solution. A 1N HCL solution shall be prepared by mixing one volume of concentrated reagent grade hydrochloric acid (38 percent) with 11 volumes of distilled water. Standardize the solution with primary standard grade, anhydrous, sodium carbonate using methyl orange as an indicator.

4.5.4.3 Test procedure. Pipette 10 mL of the sample solution (see 4.5.1.1 and 4.5.1.2) into a 250 mL conical flask and add 10 mL distilled water, 1 g of sodium cyanide and 16 drops of LaMotte sulfo-orange indicator. Titrate with the standardized 1N HCL solution to the identical color of the blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.

4.5.6.2 Test procedure. Dissolve approximately 10 g, (weighed to the third decimal), of the corrosion removing compound with water and dilute to volume in a 500 mL volumetric flask with water. Pipette 1 mL of the thoroughly-mixed solution to a 2 mL vial. Add six drops of concentrated HCL, and evaporate to loss of chlorine odor at 149°F (65°C) with a gentle flow of nitrogen. Pipette 1 mL of the internal standard solution (see 4.5.6.1.2) into the vial. Close the vial with a teflon-coated septum and heat at 149-158°F (65-70°C) for 30 minutes. After settling, decant the liquid into a second vial, and evaporate at 140°F (60°C) under a flow of nitrogen. Close the vial and, with a syringe, add 500 µL of the silylating solution (see 4.5.6.1.1) through the septum. Heat for 20 minutes at 149-158°F (65-70°C). Inject 1µL of the solution into a gas-liquid chromatograph equipped with a 6 foot X 1/8 inch stainless steel column packed with 5 percent silicone OV-225 on 80-100 mesh chromosorb W-HP. Allow the chromatogram to develop for 30 minutes under the following conditions:

Injection port	446°F (230°C)
Column oven	329°F (165°C)
Flame ionization detector	572°F (300°C)
Carrier (helium) flow	60 mL per minute

Calculate the percentage of sodium gluconate in the corrosion removing compound as follows:

$$\text{Sodium gluconate} = 500 \frac{A_X W_{IS} P}{A_{IS} W_S}$$

where A_X = area of the gluconic acid derivative peak.
A_{IS} = area of the L-ascorbic acid derivative peak.
W_{IS} = weight of L-ascorbic acid per 100 mL of solution.
W_S = weight of sample of corrosion removing compound per 500 mL.
P = correction factor determined with known concentrations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.

4.5.9 Chelating or sequestering agent content (type II only).

4.5.9.1 Preparation of ammonium oxalate indicator solution. Dissolve 5 g of reagent grade ammonium oxalate in 50 mL of distilled water in a 100 mL volumetric flask. Fill to the mark with distilled water and mix well.

4.5.9.2 Preparation of 0.2M calcium chloride solution. Dissolve exactly 22.2 g of dried reagent grade calcium chloride in distilled water in a 1 L volumetric flask. Dilute to exactly 1 L with distilled water.

4.5.9.3 Preparation of tetrasodium salt of ethylenediamine tetra-acetic acid dihydrate solution. Prepare a standard 90 g per liter solution of EDTA tetrasodium salt dihydrate as follows: Dissolve exactly 80.420 g of reagent grade ethylenediamine tetra-acetic acid disodium dihydrate in 2N sodium hydroxide solution in a 1 L volumetric flask. Fill exactly to the mark with 2N sodium hydroxide and mix well. Titrate a 50 mL portion of this solution with the calcium chloride solution (see 4.5.9.2) and record the mL of $CaCl_2$ required.

4.5.9.4 Test procedure. Pipette 50 mL of the sample solution (see 4.5.1.2) into a 300 mL conical flask. Add 150 mL of distilled water and 10 mL of the ammonium oxalate indicator solution (4.5.9.1). Titrate with the standard $CaCl_2$ solution (4.5.9.2) to the first faint turbidity which lasts for at least 30 seconds. Calculate the chelating or sequestering agent content by the following formula:

$$X = 25 \frac{V}{S}$$

Where:

X = percent chelating or sequestering agent.
V = mL of 0.2M $CaCl_2$ solution required for titration of the sample.
S = mL of the 0.2M $CaCl_2$ solution required for titration of the EDTA solution (4.5.9.3).

4.5.10 Etching characteristics (type II only). The panels used in the test for foaming characteristics (4.5.3) shall be rinsed with water, dried and examined for any evidence of etching. Check for compliance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. The corrosion removing compound shall be packaged in 55-gallon steel drums conforming to type IV of PPP-D-729. The volume of the drum shall not exceed the volume of the contents by more than 5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.

6.1.1 Type I compound. Type I compound is to be used at a concentration of 5 pounds per gallon of water. Type I corrosion removing compound contains no cyanide and no electrolytic action is needed. However, the removal of surface deposits is slow and a long soaking period is required to do a satisfactory job. This type of compound is most effective when used at a rolling boil. Agitation of the bath reduces the soaking time.

6.1.2 Type II compound. Type II compound is to be used at a concentration of 2 to 3 pounds per gallon of water. Type II solution uses both cyanide and electrolytic action. The removal of surface deposits is more rapid than with Type I compound and the soaking period is shortened. The use of type II compound is recommended where a source of direct current and methods for cyanide disposal are available. In order to reduce the loss of cyanide, the type II solution should not be used at a bath temperature higher than 130°F (54°C).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of compound (see 1.2).
(c) Level of packaging required (see section 5).
(d) Size and type of container if Level C is specified (see 5.1.2).

6.3 Definitions. The following definitions apply to terms used in this specification:

Chelate compound - Pertaining to or designating a compound which by means of 2 valences (principal, residual, or both) attaches itself to a central metallic atom so as to form a chelate group or ring.

Sequestrant compound - Sequestering agents function in the same manner as chelating compounds except that they do not form a ring.

Military custodians:
Army - MR
Navy - AS
Air Force - 68

Review activities:
Army - MI
Navy - SA, SH
Air Force - 11

User activities:
Army - AR

Agent:
Army - ME

Preparing activity:
Army - MR

Project Number 6850-0627

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:19Z
Check the source to verify that this is the current version before use.